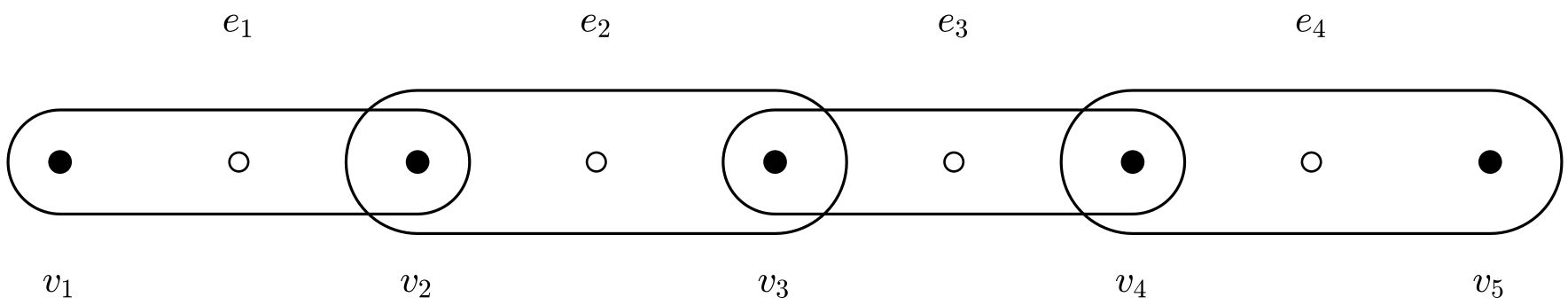
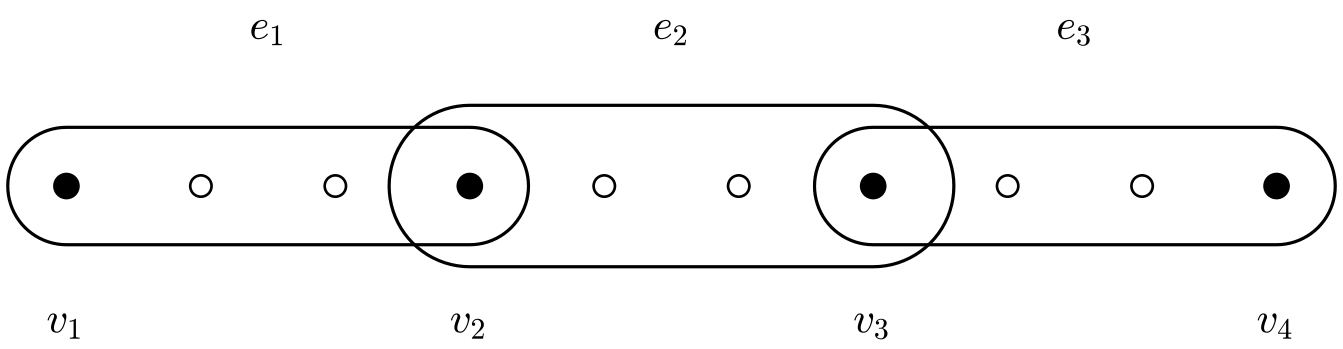
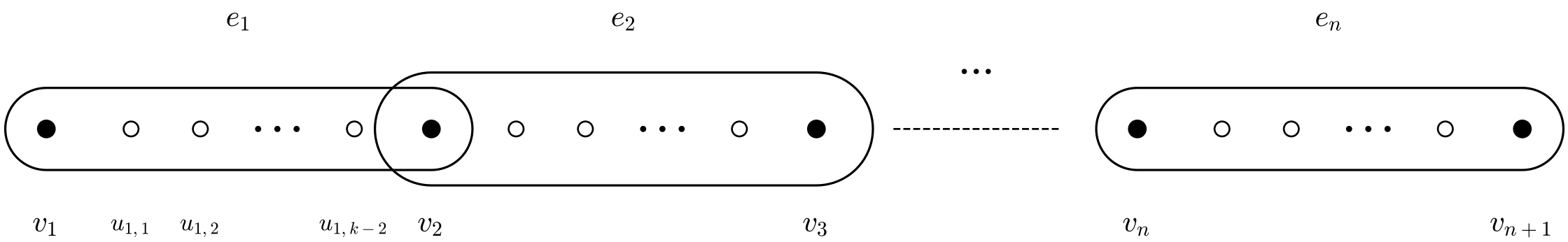
(a) $P_4^{(3)}$: $|V| = 4 \cdot 2 + 1 = 9,\ |E| = 4$



(b) $P_3^{(4)}$: $|V| = 3 \cdot 3 + 1 = 10,\ |E| = 3$



(c) $P_n^{(k)}$: $|V| = n(k-1) + 1,\ |E| = n$



$e_i = \{v_i,\ u_{i,1}, \ldots, u_{i,k-2},\ v_{i+1}\}$; $|e_i \cap e_{i+1}| = 1,\ e_i \cap e_j = \emptyset$ for $|i-j| \geq 2$